

Git Branch Merging Documentation

Project Setup:

```
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ ls
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git init
Reinitialized existing Git repository in /home/GHOST/DevOps_Aimore/employee-management-system/.git/
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "# Employee Management System" > README.md
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git add README.md
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "Initial commit: project scaffold"
[main (root-commit) 164bee4] Initial commit: project scaffold
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ ls
README.md
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ cat README.md
# Employee Management System
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git status
On branch main
nothing to commit, working tree clean
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git log --oneline --graph --decorate --all
* 164bee4 (HEAD -> main) Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ |
```

1. Fast-Forward Merge

Commands Used:

```
git checkout -b feature/Login
git add login.js
git commit -m "Add Login function skeleton"
git checkout main
git merge feature/Login
```

Description:

Moves the target branch pointer forward without creating a merge commit. This merge occurs only when the target branch has no new commits since the feature branch was created. It results in a clean and linear commit history.

```
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git branch
* main
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git status
On branch main
nothing to commit, working tree clean
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ cat README.md
# Employee Management System
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git switch -c feature/login
Switched to a new branch 'feature/login'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git branch
* feature/login
  main
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "function login() { /* auth logic */ }" > login.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ cat login.js
function login() { /* auth logic */ }
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git add login.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "Add login function skeleton"
[feature/login 8b47f0a] Add login function skeleton
 1 file changed, 1 insertion(+)
 create mode 100644 login.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git status
On branch feature/login
nothing to commit, working tree clean
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "function validateCredentials() { /* validation */ }" >> login.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ cat login.js
function login() { /* auth logic */ }
function validateCredentials() { /* validation */ }
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git status
On branch feature/login
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   login.js

no changes added to commit (use "git add" and/or "git commit -a")
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -a -m "Add credential validation"
[feature/login 83f485a] Add credential validation
 1 file changed, 1 insertion(+)
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git status
On branch feature/login
nothing to commit, working tree clean
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git switch main
Switched to branch 'main'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git merge feature/login
Updating 164bee4..83f485a
Fast-forward
 login.js | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 login.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git log --oneline --graph --decorate --all
* 83f485a (HEAD -> main, feature/login) Add credential validation
* 8b47f0a Add login function skeleton
* 164bee4 Initial commit: project scaffold
```

2. Three-Way Merge (Normal Merge)

Commands Used:

```
git switch -c feature/dashboard
git add dashboard.js
git commit -m "Add dashboard skeleton"

git checkout main
echo "## Setup Instructions" >> README.md
git add README.md
git commit -m "Add setup instructions to README"

git merge feature/dashboard -m "Merge feature/dashboard into main"
```

Description:

Creates a new merge commit because both the target branch and the feature branch contain new commits. This is the default merge strategy used when multiple developers make parallel changes.

```
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git switch -c feature/dashboard
Switched to a new branch 'feature/dashboard'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "function renderDashboard() {}" > dashboard.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ cat dashboard.js
function renderDashboard() {}
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git add dashboard.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "Add dashboard skeleton"
[feature/dashboard 3b05970] Add dashboard skeleton
 2 files changed, 2 insertions(+)
 create mode 100644 dashboard.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ glog
* 3b05970 (HEAD -> feature/dashboard) Add dashboard skeleton
* 83f485a (main, feature/login) Add credential validation
* 8b47f0a Add login function skeleton
* 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git checkout main
Switched to branch 'main'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ glog
* 3b05970 (feature/dashboard) Add dashboard skeleton
* 83f485a (HEAD -> main, feature/login) Add credential validation
* 8b47f0a Add login function skeleton
* 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "## Setup Instructions" >> README.md
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git add README.md
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "Add setup instructions to README"
[main 5d17e5f] Add setup instructions to README
 1 file changed, 1 insertion(+)
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ glog
* 5d17e5f (HEAD -> main) Add setup instructions to README
* 3b05970 (feature/dashboard) Add dashboard skeleton
|/
* 83f485a (feature/login) Add credential validation
* 8b47f0a Add login function skeleton
* 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git merge feature/dashboard -m "Merge feature/dashboard into main"
Merge made by the 'ort' strategy.
 dashboard.js | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 dashboard.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ glog
* 97e8630 (HEAD -> main) Merge feature/dashboard into main
|/
* 3b05970 (feature/dashboard) Add dashboard skeleton
* | 5d17e5f Add setup instructions to README
|/
* 83f485a (feature/login) Add credential validation
* 8b47f0a Add login function skeleton
* 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ |
```

3. Squash Merge

Commands Used:

```
git checkout -b feature/employee-module
```

```
git add employee.js
```

```
git commit -m "wip: employee class"
```

```
git checkout main
```

```
git merge --squash feature/employee-module
```

```
git status
```

```
git commit -m "Add employee module: create, add, remove, list employees"
```

Description:

Combines all commits from the feature branch into a single commit before merging into the target branch. This helps maintain a clean and readable commit history.

```
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git log --oneline --graph --decorate --all
* 97e8630 (HEAD -> main) Merge feature/dashboard into main
| \
| * 3b05970 (feature/dashboard) Add dashboard skeleton
* | 5d17e5f Add setup instructions to README
|/
* 83f485a (feature/login) Add credential validation
* 8b47f0a Add login function skeleton
* 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git branch
  feature/dashboard
  feature/login
  * main
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git switch -c feature/employee-module
Switched to a new branch 'feature/employee-module'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "class Employee {}" > employee.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git add employee.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "wip: employee class"
[feature/employee-module 55c15bd] wip: employee class
 1 file changed, 1 insertion(+)
 create mode 100644 employee.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "  addEmployee() {}" >> employee.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git add employee.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "wip: add method"
> ^C
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "wip: add method"
[feature/employee-module 14dc221] wip: add method
 1 file changed, 1 insertion(+)
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "  removeEmployee() {}" >> employee.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git add employee.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "fix typo"
[feature/employee-module ba31c70] fix typo
 1 file changed, 1 insertion(+)
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "  listEmployees() {}" >> employee.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git add employee.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "wip again"
[feature/employee-module a17763f] wip again
 1 file changed, 1 insertion(+)
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ glog
* a17763f (HEAD -> feature/employee-module) wip again
* ba31c70 fix typo
* 14dc221 wip: add method
* 55c15bd wip: employee class
* 97e8630 (main) Merge feature/dashboard into main
| \
| * 3b05970 (feature/dashboard) Add dashboard skeleton
* | 5d17e5f Add setup instructions to README
|/
* 83f485a (feature/login) Add credential validation
* 8b47f0a Add login function skeleton
* 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git checkout main
```

```

GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git checkout main
Switched to branch 'main'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ glog
* a17763f (feature/employee-module) wip again
* ba31c70 fix typo
* 14dc221 wip: add method
* 55c15bd wip: employee class
* 97e8630 (HEAD -> main) Merge feature/dashboard into main
|
| * 3b05970 (feature/dashboard) Add dashboard skeleton
* | 5d17e5f Add setup instructions to README
|/
* 83f485a (feature/login) Add credential validation
* 8b47f0a Add login function skeleton
* 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git merge --squash feature/employee-module
Updating 97e8630..a17763f
Fast-forward
Squash commit -- not updating HEAD
employee.js | 4 ++++
1 file changed, 4 insertions(+)
create mode 100644 employee.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   employee.js

GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "Add employee module: create, add, remove, list employees"
[main 9fb93f4] Add employee module: create, add, remove, list employees
1 file changed, 4 insertions(+)
create mode 100644 employee.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ glog
* 9fb93f4 (HEAD -> main) Add employee module: create, add, remove, list employees
* a17763f (feature/employee-module) wip again
* ba31c70 fix typo
* 14dc221 wip: add method
* 55c15bd wip: employee class
|
* 97e8630 Merge feature/dashboard into main
|
| * 3b05970 (feature/dashboard) Add dashboard skeleton
* | 5d17e5f Add setup instructions to README
|/
* 83f485a (feature/login) Add credential validation
* 8b47f0a Add login function skeleton
* 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ |

```

4. Rebase Merge (Rebase Strategy)

Commands Used:

```

git checkout -b feature/reports
git add reports.js
git commit -m "Add report generation skeleton"
git checkout main
git add README.md
git commit -m "Note reports module in README"
git checkout feature/reports
git rebase main

```

Description:

Replays all commits from the feature branch on top of the latest commits in the main branch. This produces a linear commit history without creating a merge commit.

```

GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git checkout -b feature/reports
Switched to a new branch 'feature/reports'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "function generateReport() {}" > reports.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git add reports.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "Add report generation skeleton"
[feature/reports c32628e] Add report generation skeleton
1 file changed, 1 insertion(+)
create mode 100644 reports.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git checkout main
Switched to branch 'main'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "## Reports Module" >> README.md
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git add README.md
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "Note reports module in README"
>
[main a934340] Note reports module in README
1 file changed, 1 insertion(+)
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ glog
* a934340 (HEAD -> main) Note reports module in README
| * c32628e (feature/reports) Add report generation skeleton
|/
| * ffbfb67 Merge branch 'feature/search'
|/
| * a5f8518 (feature/search) Set app name for search testing
* | 7379796 Rename app for enterprise release
|/
* 899226a Set app name for production
* 9fb93f4 Add employee module: create, add, remove, list employees
| * a17763f (feature/employee-module) wip again
| * ba31c70 fix typo
| * 14dc221 wip: add method
| * 55c15bd wip: employee class
|/
* 97e8630 Merge feature/dashboard into main
|/
| * 3b05970 (feature/dashboard) Add dashboard skeleton
* | 5d17e5f Add setup instructions to README
|/
* 83f485a (feature/login) Add credential validation
* 8b47f0a Add login function skeleton
* 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git checkout feature/reports
Switched to branch 'feature/reports'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ glog
* a934340 (main) Note reports module in README
| * c32628e (HEAD -> feature/reports) Add report generation skeleton
|/
| * ffbfb67 Merge branch 'feature/search'
|/
| * a5f8518 (feature/search) Set app name for search testing
* | 7379796 Rename app for enterprise release
|/
* 899226a Set app name for production
* 9fb93f4 Add employee module: create, add, remove, list employees

```

```

GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git rebase main
Successfully rebased and updated refs/heads/feature/reports.
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ glog
* 77d6aff (HEAD -> feature/reports) Add report generation skeleton
* a934340 (main) Note reports module in README
* ffbfb67 Merge branch 'feature/search'
|/
| * a5f8518 (feature/search) Set app name for search testing
* | 7379796 Rename app for enterprise release
|/
* 899226a Set app name for production
* 9fb93f4 Add employee module: create, add, remove, list employees
| * a17763f (feature/employee-module) wip again
| * ba31c70 fix typo
| * 14dc221 wip: add method
| * 55c15bd wip: employee class
|/
* 97e8630 Merge feature/dashboard into main
|/
| * 3b05970 (feature/dashboard) Add dashboard skeleton
* | 5d17e5f Add setup instructions to README
|/
* 83f485a (feature/login) Add credential validation
* 8b47f0a Add login function skeleton
* 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ |

```

5. No Fast-Forward Merge (--no-ff Merge)

Commands Used:

```
git checkout main
git checkout -b feature/settings
echo "function saveSettings() {}" > settings.js
git add settings.js
git commit -m "Add settings save function"
git checkout main
git merge --no-ff feature/settings -m "Merge feature/settings into main"
```

Description:

Forces Git to create a merge commit even when a Fast-Forward Merge is possible. This preserves the complete history of the feature branch and makes it easier to identify when a feature was merged.

```
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git checkout main
Switched to branch 'main'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git checkout -b feature/settings
Switched to a new branch 'feature/settings'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ echo "function saveSettings() {}" > settings.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git add settings.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git commit -m "Add settings save function"
[feature/settings 5b09c3b] Add settings save function
1 file changed, 1 insertion(+)
 create mode 100644 settings.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git checkout main
Switched to branch 'main'
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git merge --no-ff feature/settings -m "Merge feature/settings into main"
Merge made by the 'ort' strategy.
 settings.js | 1 +
1 file changed, 1 insertion(+)
 create mode 100644 settings.js
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ glog
* f7b8924 (HEAD -> main) Merge feature/settings into main
|
| * 5b09c3b (feature/settings) Add settings save function
|
| * 77d6aff (feature/reports) Add report generation skeleton
|
| * a934340 Note reports module in README
| * ffbfb67 Merge branch 'feature/search'
|
| * a5f8518 (feature/search) Set app name for search testing
| * 7379796 Rename app for enterprise release
|
| * 899226a Set app name for production
| * 9fb93f4 Add employee module: create, add, remove, list employees
| * a17763f (feature/employee-module) wip again
| * ba31c70 fix typo
| * 14dc221 wip: add method
| * 55c15bd wip: employee class
|
| * 97e8630 Merge feature/dashboard into main
|
| * 3b05970 (feature/dashboard) Add dashboard skeleton
| * 5d17e5f Add setup instructions to README
|
| * 83f485a (feature/login) Add credential validation
| * 8b47f0a Add login function skeleton
| * 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ |
```

Final Output:

```
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ tree
.
├── README.md
├── config.txt
├── dashboard.js
├── employee.js
├── login.js
└── settings.js

1 directory, 6 files
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git status
On branch main
nothing to commit, working tree clean
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git branch
  feature/dashboard
  feature/employee-module
  feature/login
  feature/reports
  feature/search
  feature/settings
* main
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ git log --oneline --graph --decorate --all
*   f7b8924 (HEAD -> main) Merge feature/settings into main
/\
 * 5b09c3b (feature/settings) Add settings save function
 * 77d6aff (feature/reports) Add report generation skeleton
/\
 * a934340 Note reports module in README
 * ffbfb67 Merge branch 'feature/search'
/\
 * a5f8518 (feature/search) Set app name for search testing
 * 7379796 Rename app for enterprise release
/\
 * 899226a Set app name for production
 * 9fb93f4 Add employee module: create, add, remove, list employees
 * a17763f (feature/employee-module) wip again
 * ba31c70 fix typo
 * 14dc221 wip: add method
 * 55c15bd wip: employee class
/\
 * 97e8630 Merge feature/dashboard into main
/\
 * 3b05970 (feature/dashboard) Add dashboard skeleton
 * | 5d17e5f Add setup instructions to README
|/
 * 83f485a (feature/login) Add credential validation
 * 8b47f0a Add login function skeleton
 * 164bee4 Initial commit: project scaffold
GHOST@GHOST:~/DevOps_Aimore/employee-management-system$ |
```